

Cohort Study Analysis

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1 Introduction

The aim of this analysis is to investigate mortality over time among workers employed at the Porto Marghera petrochemical plant. The data refer to a retrospective cohort.

The beginning and end of follow-up were set at 1973 and 1999, respectively. This is a dynamic-population study, since subjects may enter follow-up at different times.

The association under investigation is between the number of years elapsed since the beginning of employment—defined as entry into risk exposure—and death.

2 Empirical Strategy

The dichotomous variable `EVENTO` was created to indicate whether a subject experienced death.

The variables `Eta_esp`, `Eta_start`, and `Eta_end` were generated to represent, respectively, age at the beginning of exposure, age at the beginning of follow-up, and age at the end of follow-up (if the event was not observed) or age at death (if the event occurred).

The variables `Durata_fup`, `Latenza`, and `dur_lav_start` were generated to represent, respectively: the time from the beginning of follow-up to the event, if observed, or to the end of follow-up, if the event was not observed (in which case the subject was right-censored); the duration of exposure, which may also be interpreted as length of employment; and the time from the start of exposure (beginning of employment) to the beginning of follow-up.

The variable `ingresso` was generated to classify the year in which exposure began. The categories were: < 1969 , 1969–1972, 1973–1984, and ≥ 1985 .

The `stset` command was used to declare the data as survival-time data. Time begins in the first year of exposure; the time at which a subject starts being observed is set to the beginning of follow-up; and the end time is set to the end of follow-up, or to the year of death if the event occurred.

```
stset yrout, failure(EVENTO) origin(time yrin) enter(time yrin73) id(idsogget)
```

The data were subsequently split by calendar period (1973–1982, 1983–1999, and > 2000), by age using five-year intervals, and by latency using ten-year intervals. Three time-dependent variables—`cal`, `eta`, and `lat`—were therefore created.

```
stsplint cal=0, at(1973, 1983, 1993, 2000)
stsplint eta, after(time=ynasc) at(15(5)90)
stsplint lat, after(time=yrin) at(0,10,20,30,40,50)
```

After splitting the data, the additional time-dependent variables `ap`, `anno_corr`, and `eta_corr` were generated. These denote person-years (calculated as the difference between exit time `_t` and entry time `_t0`), current calendar year, and current age, respectively.

The overall rate and crude rates for each stratum were calculated using the formulas in Appendix A.1. Mantel–Haenszel comparisons (Appendix A.2) were then carried out for the latency categories.

Stratification was used to assess confounding. Estimates were adjusted for age and calendar period as potential confounders.

```
strate
strate lat
stmh lat, compare(10,0)
stmh lat cal eta, compare(10,0)
```

The data were aggregated by strata of `cal`, `lat`, and `eta`. In this format, crude and stratified rates can be calculated with the `dystate` command (Appendix A.3).

```
collapse (sum) EVENTO ap, by(cal eta lat)
```

Finally, indirect standardization was proposed as an additional method for controlling confounders. Because no external standard population was available, person-years were used as the weight for each stratum (Appendix A.4).

```
ir EVENTO lat ap, by(eta) estandard
ir EVENTO lat ap, by(cal) estandard
```

3 Results

3.1 Descriptive Statistics

The dataset contains 1,658 subjects at risk. Of these, only 248 (approximately 15%) experienced the event of interest (death).

- `yrim` (year of first exposure), mean: 1966.8;
- `yrou` (year of end of follow-up or death), mean: 1998.3;
- `ynasc` (year of birth), mean: 1937.8;
- `yrim73` (year of entry into follow-up), mean: 1973.4.

Mean values of the generated variables were as follows:

- age at first exposure: 29 years;
- age at death/end of follow-up: 60.5 years;
- mean follow-up duration: 24.8 years;
- mean latency: 31.4 years;
- mean time between exposure and follow-up: 6.6 years.

3.2 Survival-Time Data

After the data were set up for survival analysis, 248 of the 1,658 subjects were observed to have experienced the event, and no subject left the study before the end of follow-up. The cohort was followed for a total of 41,140.47 person-years.

Risk time zero corresponds to the first year of exposure. The entry time indicates that at least one subject entered follow-up in the same year as first exposure. The maximum observed survival time was approximately 49 years from exposure (Table 1).

```

Survival-time data settings

      ID variable: idsogget
      Failure event: EVENTO!=0 & EVENTO<.
Observed time interval: (yrout[_n-1], yrout]
      Enter on or after: time yrin73
      Exit on or before: failure
      Time for analysis: (time-origin)
      Origin: time yrin

```

```

1,658 total observations
  0 exclusions

```

```

1,658 observations remaining, representing
1,658 subjects
  248 failures in single-failure-per-subject data
41,140.469 total analysis time at risk and under observation
               At risk from t = 0
               Earliest observed entry t = 0
               Last observed exit t = 49.04382

```

Figure 1: stset output.

Approximately six events per 1,000 person-years were observed. The median and 75th percentile could not be estimated because fewer than 50% of subjects experienced the event (Table 2).

```

      Failure _d: EVENTO
Analysis time _t: (yrout-origin)
      Origin: time yrin
Enter on or after: time yrin73
      ID variable: idsogget

```

	Time at risk	Incidence rate	Number of subjects	Survival time		
				25%	50%	75%
Total	41,140.4685	.0060281	1658	40.82202	.	.

Figure 2: stsum output.

On average, subjects entered follow-up 6.6 years after first exposure. The mean time to the event or censoring was approximately 31.4 years, whereas mean time at risk from first exposure to the event or censoring was approximately 29 years (Table 3).

```

Failure _d: EVENT0
Analysis time _t: (yrout-origin)
Origin: time yrin
Enter on or after: time yrin73
ID variable: idsogget

```

Category	Total	Per subject			
		Mean	Min	Median	Max
Number of subjects	1658				
Number of records	1658	1	1	1	1
Entry time (first)		6.599306	0	4.876831	22.38403
Exit time (final)		31.41262	2.660522	29.45447	49.04382
Subjects with gap	0				
Time on gap	0	.	.	.	.
Time at risk	41140.469	24.81331	.0267334	26.65979	26.65979
Failures	248	.1495778	0	0	1

Figure 3: stdes output.

Subjects entered follow-up in a relatively balanced manner across the generated calendar periods (Table 4). Each episode, expressed as person-years in Table 5, represents the time spent by a subject in that calendar period.

Observation interval	Freq.	Percent	Cum.
1973	1,645	34.46	34.46
1983	1,613	33.79	68.24
1993	1,516	31.76	100.00
Total	4,774	100.00	

Figure 4: Tabulation of cal: episodes.

	Total
Observation interval	
1973	15,666.5
1983	15,699
1993	9,774.91
Total	41,140.5

Figure 5: Tabulation of cal: person-years.

The largest numbers of episodes and person-years occurred between ages 35 and 60 (Tables 6 and 7).

Observation interval	Freq.	Percent	Cum.
15	11	0.08	0.08
20	230	1.77	1.85
25	514	3.95	5.80
30	967	7.43	13.24
35	1,408	10.82	24.06
40	1,893	14.55	38.61
45	2,164	16.63	55.25
50	1,985	15.26	70.51
55	1,650	12.68	83.19
60	1,191	9.16	92.34
65	624	4.80	97.14
70	266	2.04	99.19
75	86	0.66	99.85
80	18	0.14	99.98
85	2	0.02	100.00
Total	13,009	100.00	

Figure 6: Tabulation of eta: episodes.

	Total
Observation interval	
15	5.96851
20	492.218
25	1,756.14
30	3,030.45
35	4,774.92
40	6,359.06
45	7,075.15
50	6,321.35
55	5,106.31
60	3,501.76
65	1,812.95
70	667.408
75	191.977
80	42.1166
85	2.69604
Total	41,140.5

Figure 7: Tabulation of eta: person-years.

For latency, the largest numbers of episodes and person-years were observed between 10 and 30 years (Tables 8 and 9).

Observation interval	Freq.	Percent	Cum.
0	3,036	18.28	18.28
10	5,317	32.02	50.30
20	5,405	32.55	82.84
30	2,473	14.89	97.74
40	376	2.26	100.00
Total	16,607	100.00	

Figure 8: Tabulation of lat: episodes.

	Total
Observation interval	
0	8,065.33
10	13,761.1
20	13,092.1
30	5,546.95
40	674.907
Total	41,140.5

Figure 9: Tabulation of lat: person-years.

The estimated Kaplan–Meier survival curve decreases very slowly over time. At 50 years after exposure, the estimated survival probability is approximately 70% (Figure 10).

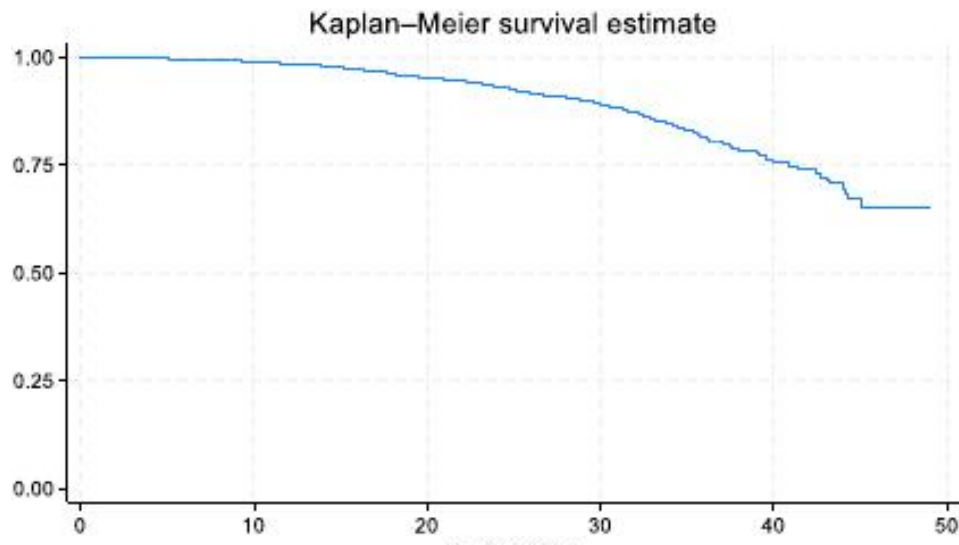


Figure 10: Overall Kaplan-Meier survival curve.

The survival curves stratified by latency category show a more pronounced increase in risk as the number of years since first exposure increases (Figure 11).

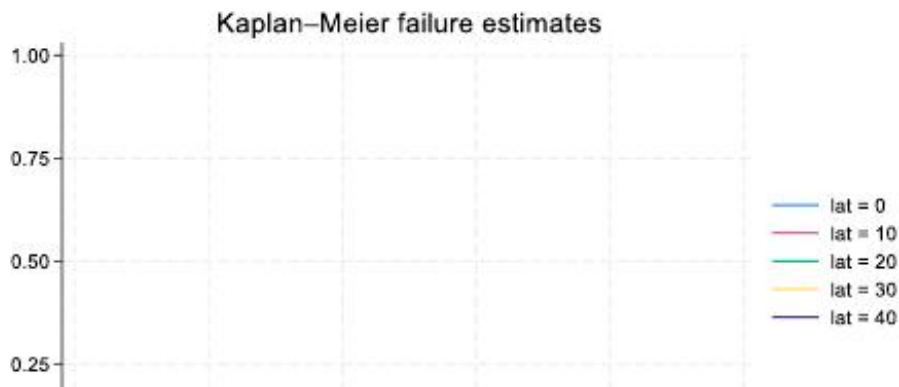


Figure 11: Kaplan-Meier failure curves by latency category.

The age-stratified curves show that the younger categories have nearly flat curves, whereas the older categories rise more rapidly. Above 75–80 years, the curves become irregular because few subjects remain, producing unstable estimates (Figure 12).

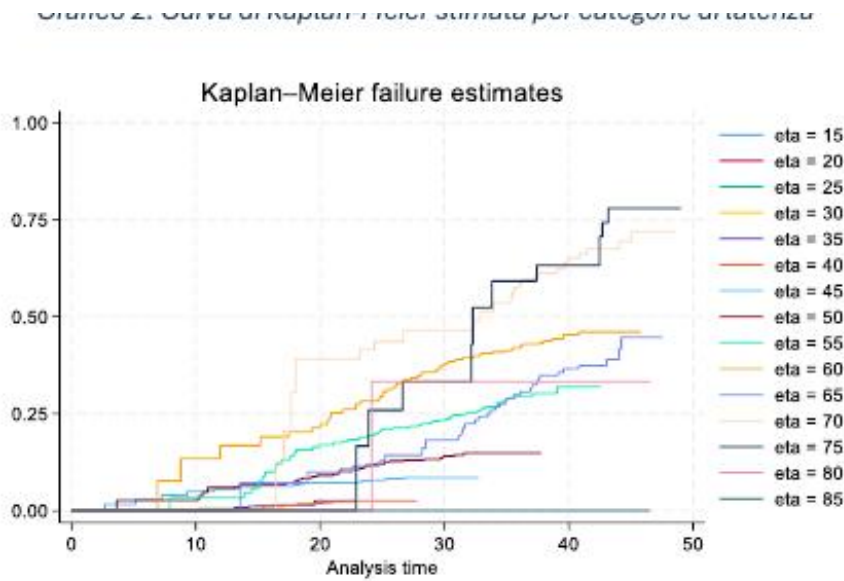


Figure 12: Kaplan-Meier failure curves by age category.

Curves for earlier calendar periods show a higher failure probability at the beginning of follow-up (Figure 13).

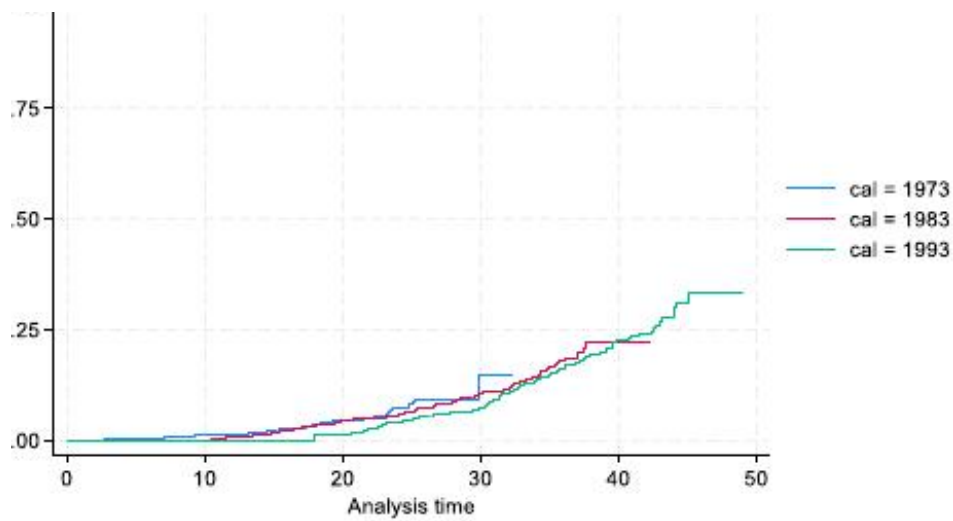


Figure 13: Kaplan-Meier failure curves by calendar period.

Curves stratified by period of entry into risk show that individuals first exposed in earlier periods had longer exposure durations, which is consistent with higher mortality. More recent cohorts show higher survival, but their follow-up is shorter. In the latter case, the loss of information due to censoring must be considered (Figure 14).

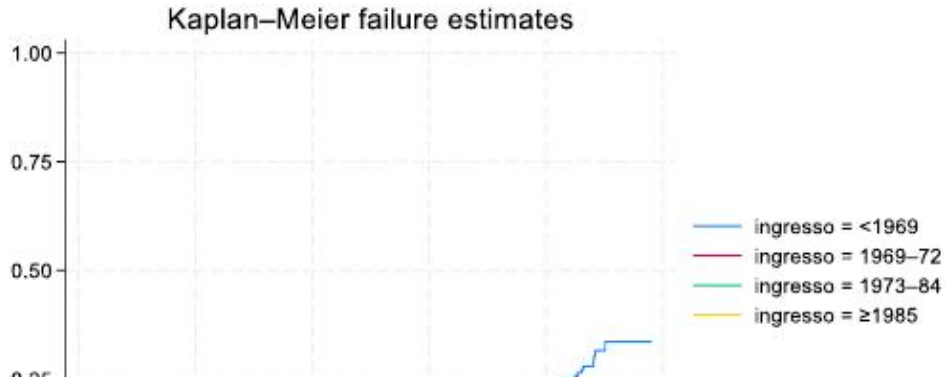


Figure 14: Kaplan-Meier curves by period of entry into risk.

3.3 Crude Rates

Crude rates describe the incidence of the event within each group without making adjusted comparisons. Examples of the calculations are provided in Appendix A.5.

The overall mortality rate was six deaths per 1,000 person-years, confirming that mortality in this cohort was low (Table 15).

D	Y	Rate	Lower	Upper
248	4.1e+04	0.0060281	0.0053227	0.0068271

Figure 15: Overall crude rate.

After stratification by calendar period, incidence increased and reached a maximum of 10.8 per 1,000 person-years in the last category (Table 16).

cal	D	Y	Rate	Lower	Upper
1973	45	1.6e+04	0.0028724	0.0021446	0.0038471
1983	97	1.6e+04	0.0061787	0.0050637	0.0075392
1993	106	9.8e+03	0.0108441	0.0089643	0.0131180

Figure 16: Crude rate by calendar period.

After stratification by age, incidence increased up to age 75, when it reached 57 deaths per 1,000 person-years. Rates were lower in the extreme age groups because of the small number of observations, as also reflected by the wider confidence intervals (Table 17).

eta	D	Y	Rate	Lower	Upper
15	0	5.9685	0.0e+00	.	.
20	0	492.2181	0.0e+00	.	.
25	0	1.8e+03	0.0e+00	.	.
30	1	3.0e+03	0.0003300	0.0000465	0.0023426
35	7	4.8e+03	0.0014660	0.0006989	0.0030751
40	11	6.4e+03	0.0017298	0.0009580	0.0031235
45	17	7.1e+03	0.0024028	0.0014937	0.0038651
50	32	6.3e+03	0.0050622	0.0035799	0.0071584
55	56	5.1e+03	0.0109668	0.0084398	0.0142504
60	54	3.5e+03	0.0154208	0.0118107	0.0201345
65	34	1.8e+03	0.0187540	0.0134003	0.0262466
70	24	667.4081	0.0359600	0.0241029	0.0536501
75	11	191.9768	0.0572986	0.0317320	0.1034644
80	1	42.1166	0.0237436	0.0033446	0.1685577
85	0	2.6960	0.0e+00	.	.

Figure 17: Crude rate by age.

After stratification by latency, incidence increased with the number of years since first exposure. At 40 years after first exposure, incidence was approximately 19 cases per 1,000 person-years (Table 18).

lat	D	Y	Rate	Lower	Upper
0	9	8.1e+03	0.0011159	0.0005806	0.0021446
10	55	1.4e+04	0.0039968	0.0030685	0.0052058
20	85	1.3e+04	0.0064924	0.0052491	0.0080303
30	86	5.5e+03	0.0155040	0.0125504	0.0191528
40	13	674.9070	0.0192619	0.0111846	0.0331727

Figure 18: Crude rate by latency.

3.4 Mantel–Haenszel Rate Ratios

The Mantel–Haenszel estimator was calculated to compare latency categories. Relative to the reference latency category and without adjustment for confounders:

- at 10 years since first exposure, incidence was approximately 3.6 times that at 0 years;
- at 20 years, incidence was 5.8 times that at 0 years;
- at 30 years, incidence was 13.9 times that at 0 years;
- at 40 years, incidence was 17.3 times that at 0 years.

These findings confirm the increasing incidence with time since exposure already observed in the crude rates.

The comparison between latency categories was repeated while treating `cal` and `eta` as confounders:

- at 10 years, incidence was 2.2 times that at 0 years (not statistically significant);
- at 20 years, incidence was 1.53 times that at 0 years (not statistically significant);
- at 30 years, incidence was 11.3 times that at 0 years (not statistically significant);
- the estimate at 40 years was not estimable.

The association between exposure and the event was attenuated after controlling for confounding. Beyond 30 years of latency, estimates became unstable because of the small sample size, producing wide confidence intervals or non-estimable risks. The reduction in the effect suggests that both age and calendar period predict mortality.

3.5 Standardized Rates

After aggregation, adjusted incidence-rate ratios were calculated by stratifying on `eta` and on `cal`.

Because observations were sparse at the extremes of the age distribution, subjects younger than 40 or older than 65 years were removed to obtain a valid age-adjusted standardization. This choice is supported by the table in Appendix A.6. The association between latency and incidence decreased to 2.23 and became non-significant (Table 19).

Observation inte	IRR	[95% conf. interval]		Weight
40	.	.484346	.	1028.659 (exact)
45	.3743911	.1156621	1.5762	730.8464 (exact)
50	2.361129	.3934959	96.22991	447.3923 (exact)
55	2.522808	.4343067	101.4344	223.9502 (exact)
60	.5544155	.1459579	4.699168	73.11133 (exact)
65	.	.0494183	.	10.20959 (exact)
Crude	2.226746	1.108068	5.230366	(exact)
E. standardized	1.225708	.5975136	2.514353	

Figure 19: Standardized rate adjusted for age.

After adjustment for calendar period (Table 20), the incidence-rate ratio was 4.06. Thus, mortality was approximately four times higher with increasing time since exposure after accounting for calendar period. Because the standardized rate differs from the crude rate, calendar period acts as a confounder.

Observation inte	IRR	[95% conf. interval]		Weight
1973	3.540635	1.673412	8.358618	7356.063 (exact)
1983	.	1.18265	.	688.1298 (exact)
1993	.	.0611832	.	21.13354 (exact)
Crude	6.475534	3.354319	14.33503	(exact)
E. standardized	4.060229	1.989941	8.284394	

Figure 20: Standardized rate adjusted for calendar period.

4 Conclusion

The analyses based on the Mantel–Haenszel estimator and the standardized incidence-rate ratio confirm the influence of confounding. Without adjustment, risk increases with the number of years elapsed since first exposure. After adjustment for confounders, the effect is attenuated and becomes unstable. This instability results from the low mortality in the cohort and the limited number of person-years in some strata, leading to wide confidence intervals and less precise estimates.

A Appendix

A.1 Notation

a_j	Cases in stratum j
b_j	Cases in the reference category of stratum j
T_{1j}	Person-years in stratum j
T_{0j}	Person-years in the reference category of stratum j
t_j	Person-years in stratum j
w_j	Weight assigned to stratum j

A.2 Formulas for Crude Rates

Overall crude rate:

$$IR = \frac{\sum_j a_j}{\sum_j t_j}.$$

Stratum-specific crude rate:

$$IR_j = \frac{a_j}{t_j}.$$

A.3 Mantel–Haenszel Rate-Ratio Formula

$$RR_{MH} = \frac{\sum_j \left(\frac{a_j T_{0j}}{t_j} \right)}{\sum_j \left(\frac{b_j T_{1j}}{t_j} \right)}.$$

A.4 Alternative Commands for Crude Rates

```
dyrate EVENT0, exposure(ap)
dyrate EVENT0 lat, exposure(ap)
dymh EVENT0 lat, exposure(ap) compare(10,0)
dymh EVENT0 lat cal eta, exposure(ap) compare(10,0)
```

A.5 Standardized-Rate Formula

$$SRR = \frac{\sum_j IR_j w_j}{\sum_j w_j}.$$

A.6 Worked Example

```
Failure _d: EVENT0
Analysis time _t: (yrout-origin)
Origin: time yrin
Enter on or after: time yrin73
ID variable: idsogget
```

lat	Time at risk	Incidence rate	Number of subjects	Survival time		
				25%	50%	75%
0	8,065.32581	.0011159	1033	.	.	.
10	13,761.1475	.0039968	1630	.	.	.
20	13,092.1398	.0064924	1577	.	.	.
30	5,546.94849	.015504	799	.	.	.
40	674.9069824	.0192619	223	.	.	.
Total	41,140.4685	.0060281	1658	40.82202	.	.

Figure 21: stsum by latency category.

EVENT0	Observation interval					Total
	0	10	20	30	40	
Vivo	0	17	693	490	210	1,410
Deceduto	9	55	85	86	13	248
Total	9	72	778	576	223	1,658

Figure 22: Cross-tabulation of EVENT0 and lat.

Overall rate:

$$\frac{248}{41140.469} = 0.00602813.$$

Rate for latency category 10:

$$\frac{55}{13761.1475} = 0.00399676.$$

Incidence for latency category 10 relative to category 0:

$$\frac{0.0039968}{0.0011159} = 3.5816829.$$

For the incidence in latency category 10 relative to category 0, adjusted for confounders, consider `stsum, by(cal eta lat)` with `cal=1973` and `eta=45`:

$$0.0026130 \times 1,148.08752 = 3 \quad (\text{rate} \times \text{person-years for latency 10}),$$

$$\text{Numerator} = 3 \times \frac{650.0567627}{1,148.08752 + 650.0567627} = 1.084,$$

that is, cases in latency stratum 10 multiplied by person-years in latency stratum 0 and divided by the combined person-years of the two strata.

$$0.0061533 \times 650.0567627 = 4 \quad (\text{rate} \times \text{person-years for latency 0}),$$

$$\text{Denominator} = 4 \times \frac{1,148.08752}{1,148.08752 + 650.0567627} = 2.533,$$

that is, cases in latency stratum 0 multiplied by person-years in latency stratum 10 and divided by the combined person-years of the two strata.

A.7 Complete Age-Stratified Table

Observation inte	IRR	[95% conf. interval]		Weight
15	.	.	.	5.968506
20	.	.	.	492.2181
25	.	.	.	1714.839 (exact)
30	0	0	69.02847	1936.407 (exact)
35	.	.5989322	.	1401.724 (exact)
40	.	.484346	.	1028.659 (exact)
45	.3743911	.1156621	1.5762	730.8464 (exact)
50	2.361129	.3934959	96.22991	447.3923 (exact)
55	2.522808	.4343067	101.4344	223.9502 (exact)
60	.5544155	.1459579	4.699168	73.11133 (exact)
65	.	.0494183	.	10.20959 (exact)
70	.	.	.	0
75	.	.	.	0
80	.	.	.	0
85	.	.	.	0
Crude	6.475534	3.354319	14.33503	(exact)
E. standardized	.	.	.	

Figure 23: Complete age-stratified table used for standardized rates.